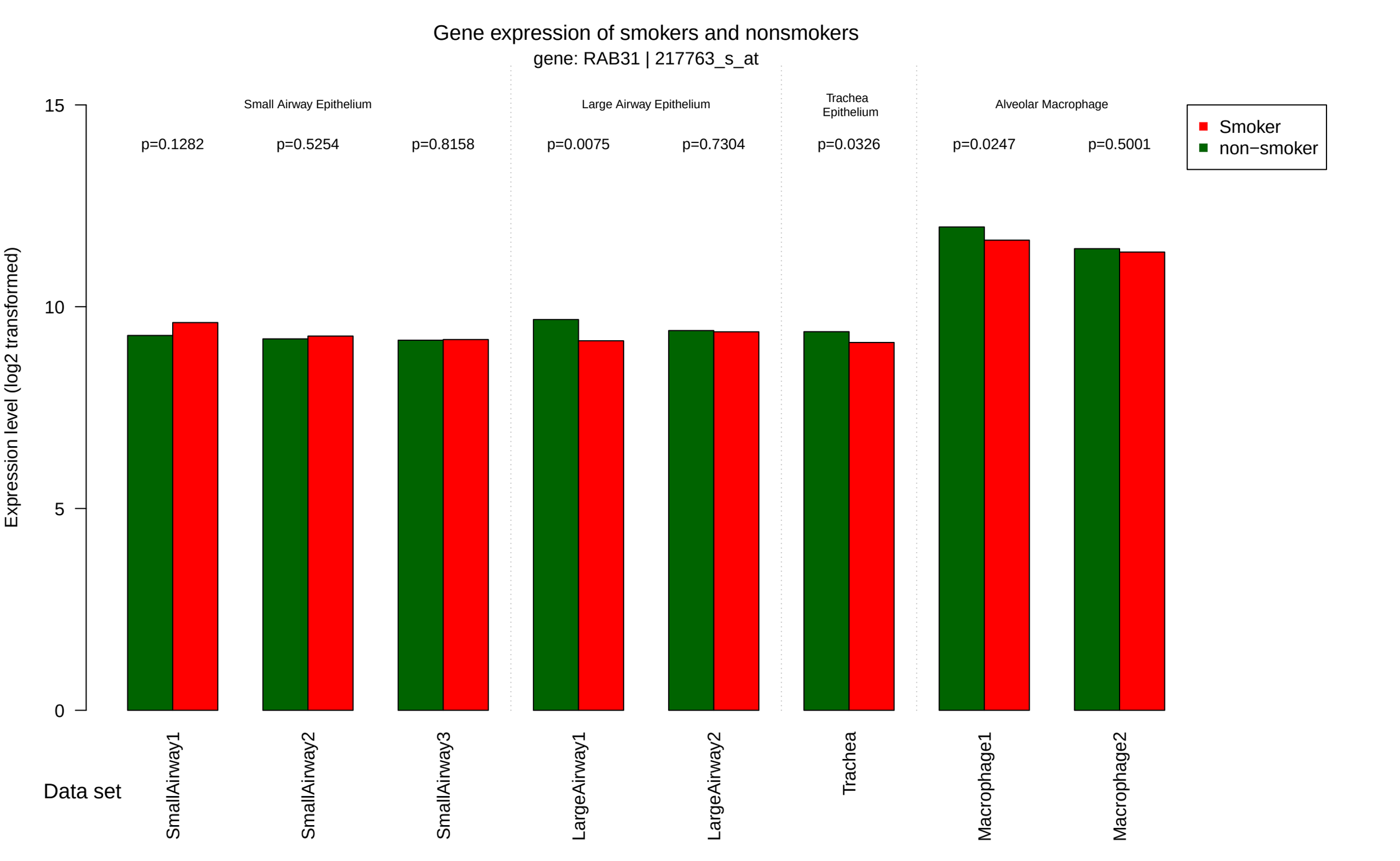
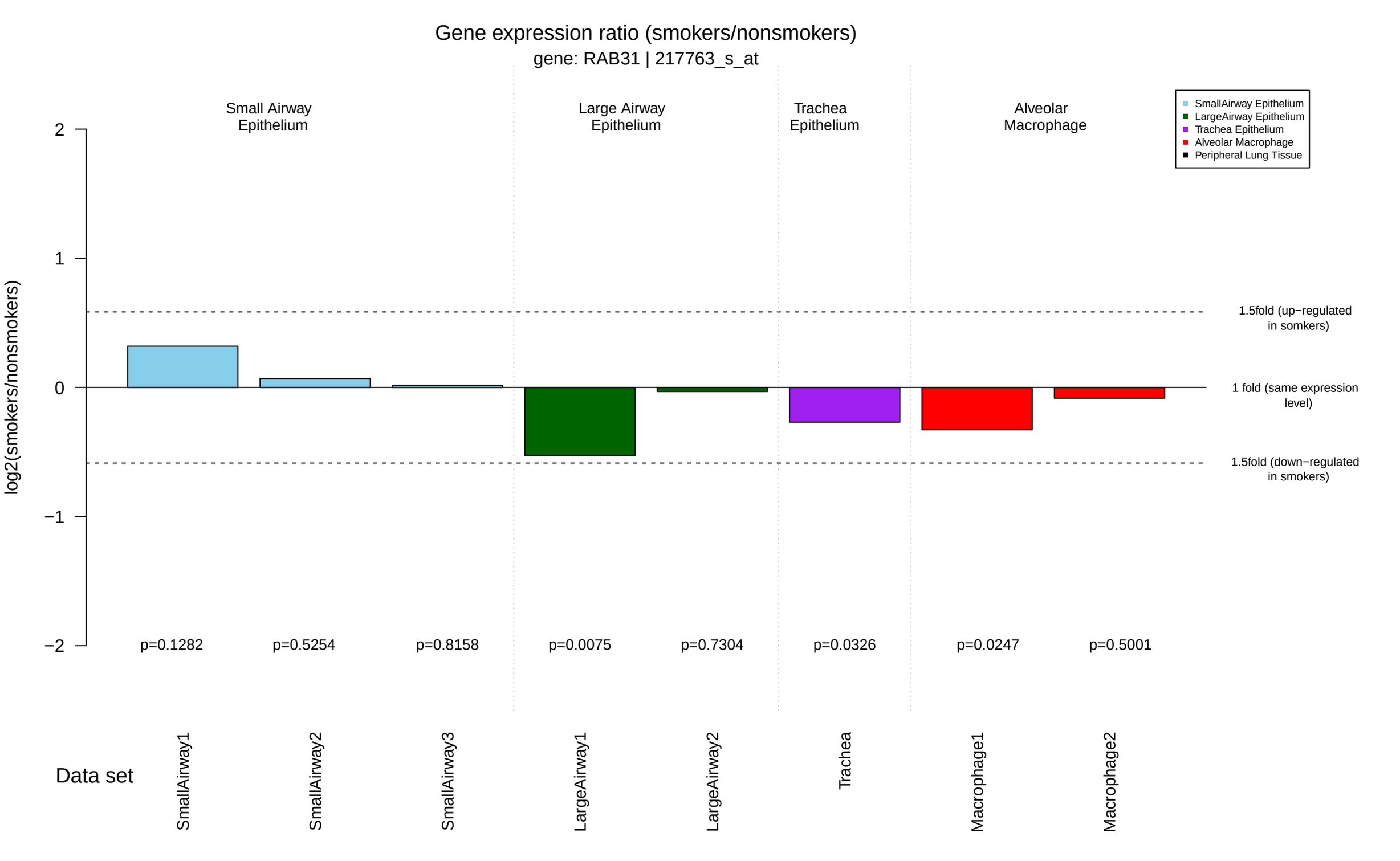
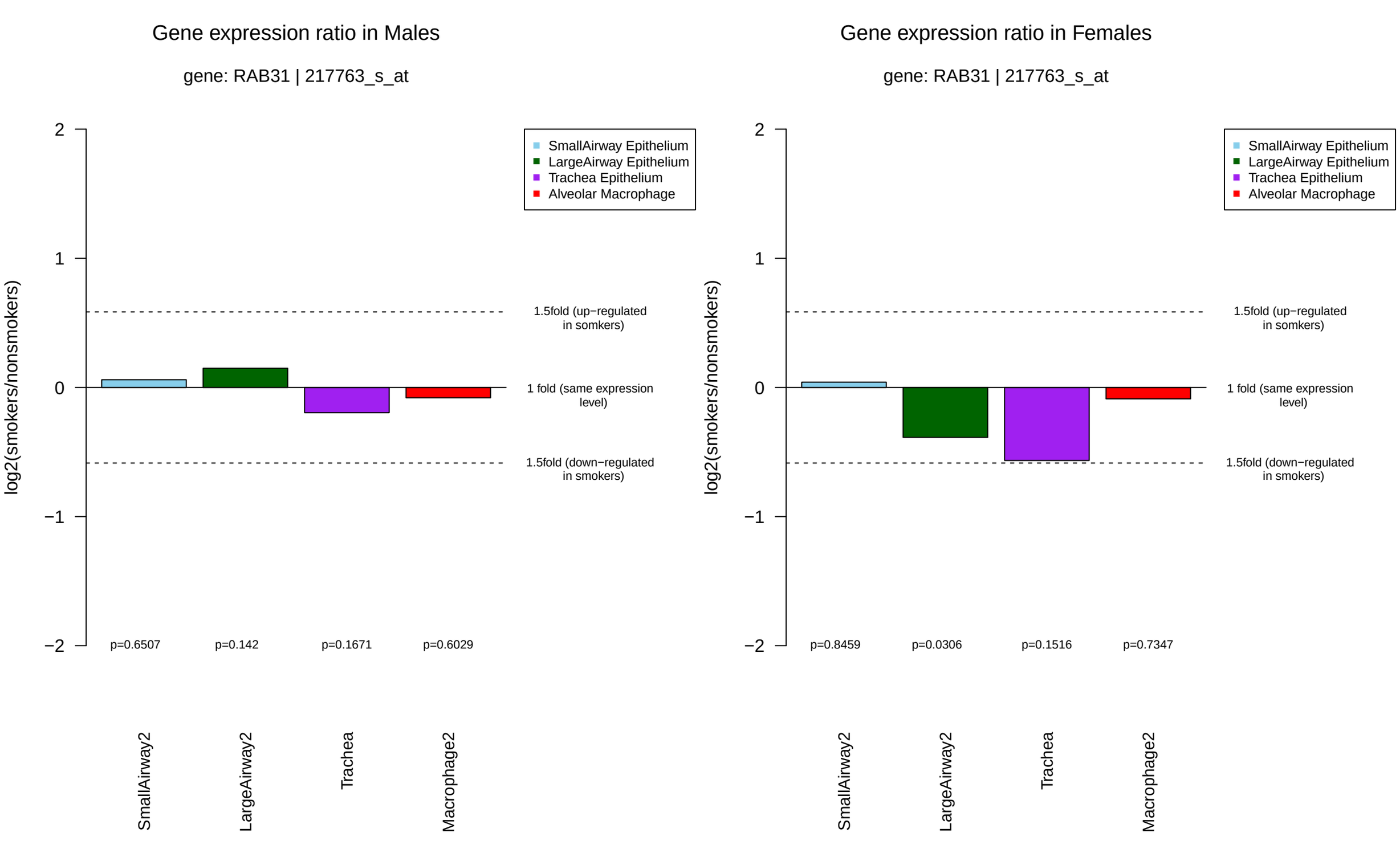
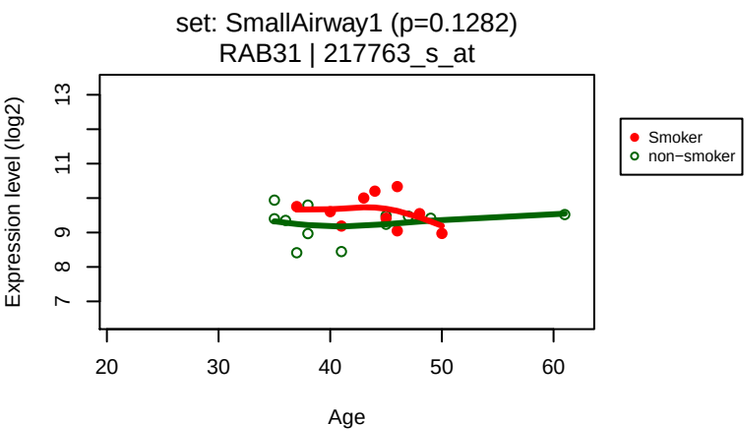
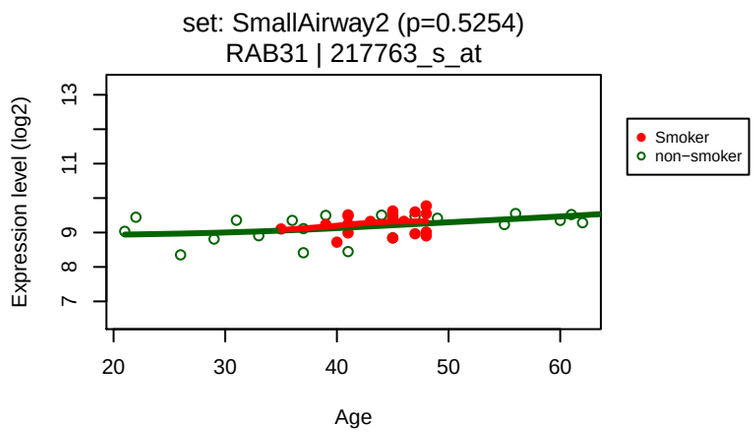
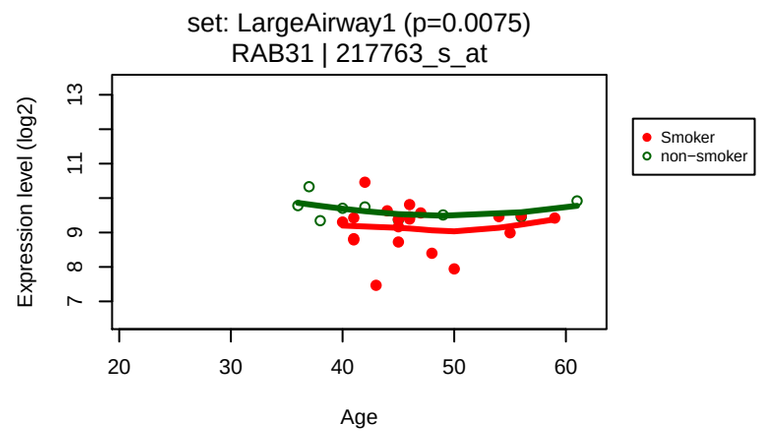

Gene expression of smokers and nonsmokers
gene: RAB31 | 217763_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1282
p=0.5254
p=0.8158
p=0.0075
p=0.7304
p=0.0326
p=0.0247
p=0.5001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

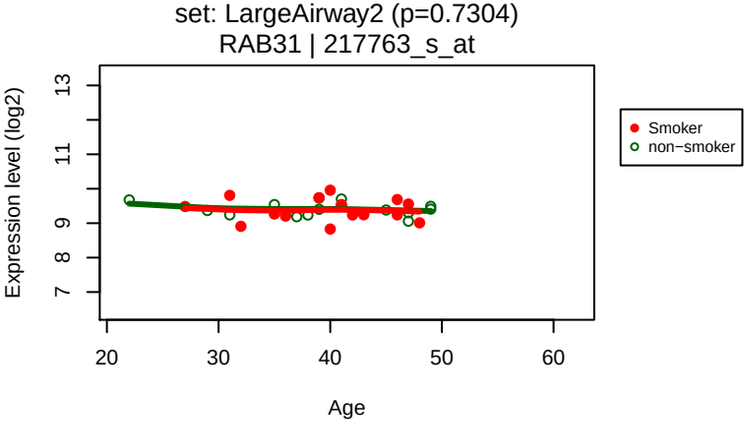

Gene expression ratio (smokers/nonsmokers)
gene: RAB31 | 217763_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1282
p=0.5254
p=0.8158
p=0.0075
p=0.7304
p=0.0326
p=0.0247
p=0.5001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

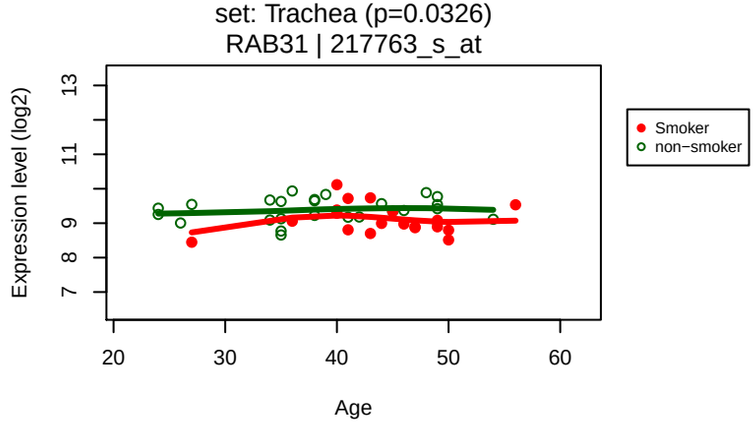

Gene expression ratio in Males
gene: RAB31 | 217763_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6507
p=0.142
p=0.1671
p=0.6029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB31 | 217763_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8459
p=0.0306
p=0.1516
p=0.7347
SmallAirway2
LargeAirway2
Trachea
Macrophage2

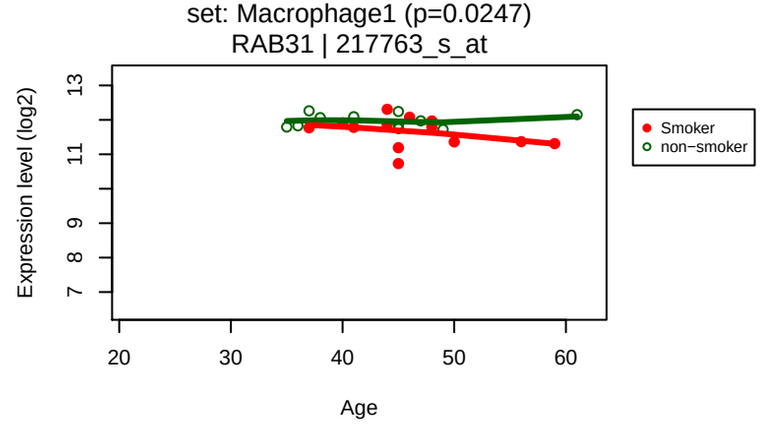

set: SmallAirway1 (p=0.1282)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker

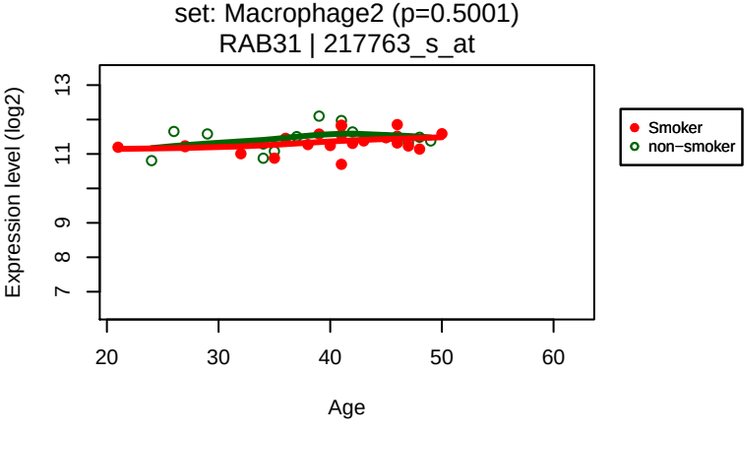

set: SmallAirway2 (p=0.5254)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker

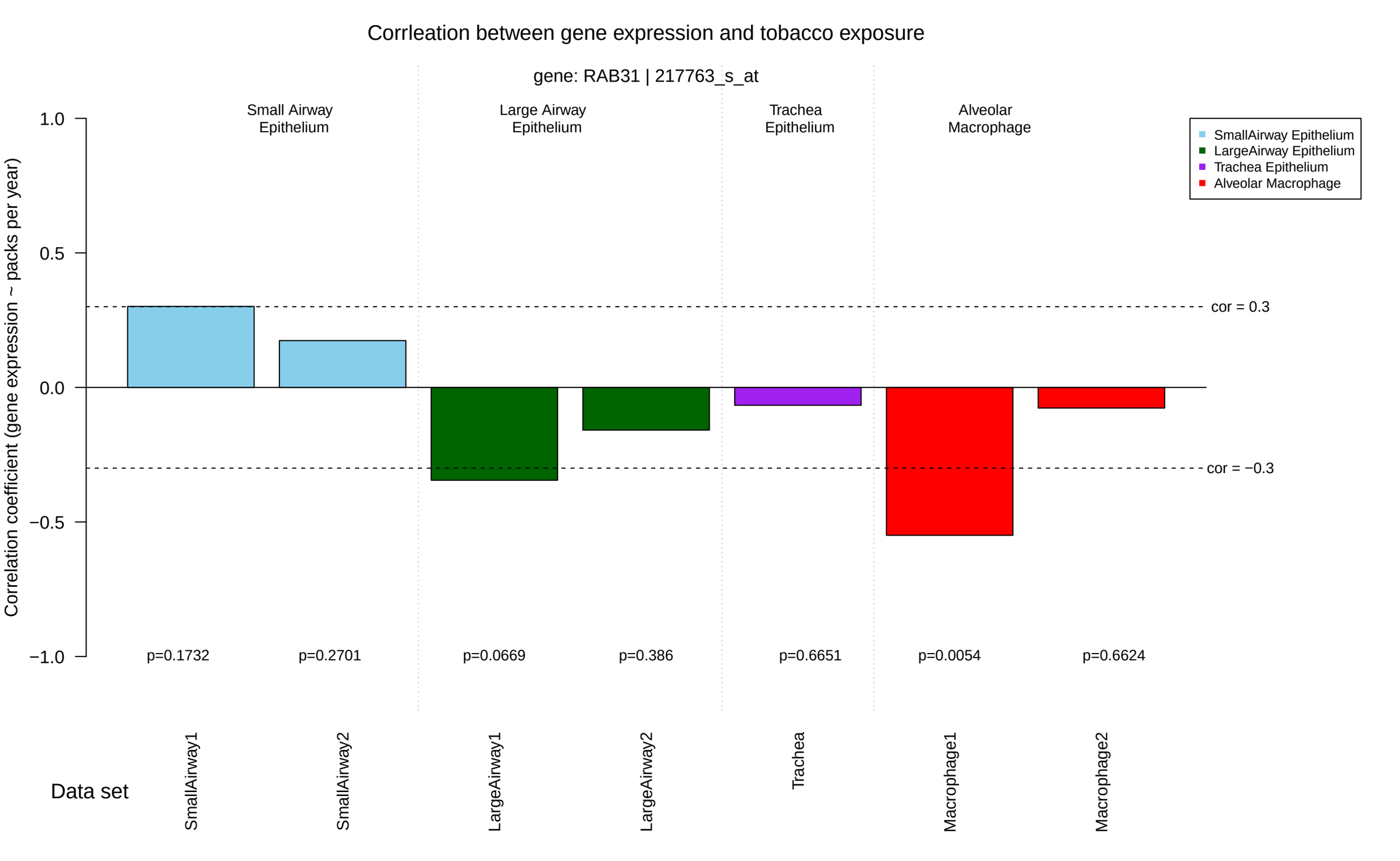

set: LargeAirway1 (p=0.0075)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7304)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0326)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0247)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5001)
RAB31 | 217763_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB31 | 217763_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1732
p=0.2701
p=0.0669
p=0.386
p=0.6651
p=0.0054
p=0.6624
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2